MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

OSCILLATOR, SQUARE WAVE, ANALOG, KEY CODE MVJ

INACTIVE FOR DESIGN AFTER DATE OF THIS SPECIFICATION

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic square wave oscillator module.

1.2 Classification. This specification covers one class of modules.

Class I as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 1.50 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be MVJ.

3.5.2 Size. The module size shall be 1C.

3.5.3 Weight. The module shall have a maximum weight of 227 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/132 -1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 25 Vdc ±20% when driving the specified loads.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

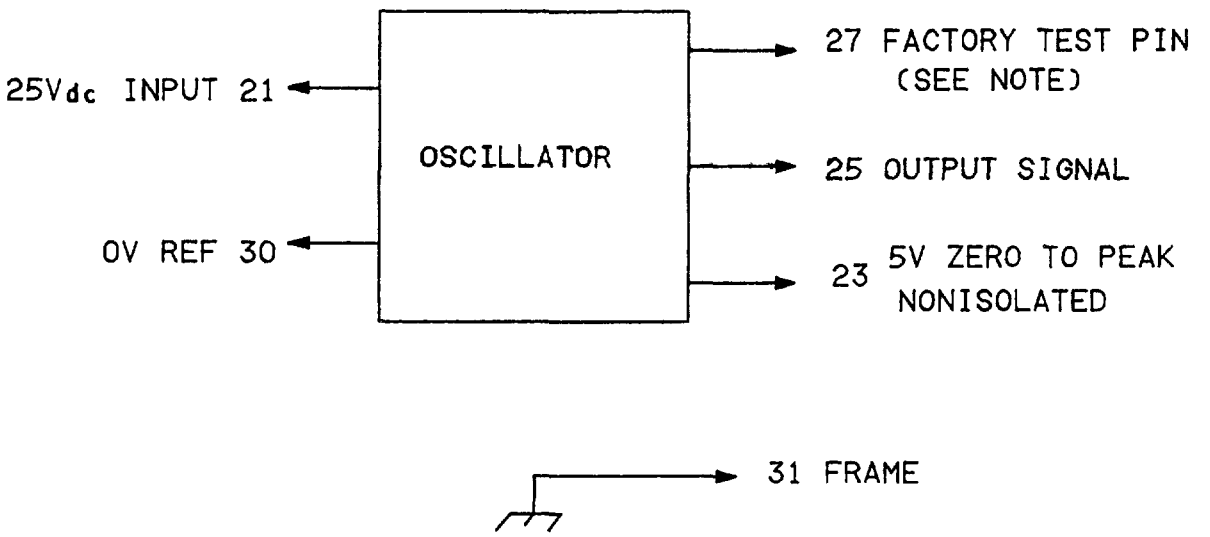
4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

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NOTE:
PIN 27 IS USED TO CALIBRATE OR TEST THE MODULE

FIGURE 1. MVJ module function diagram, oscillator, square wave.

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TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/,2/	Acceptance requirements 25°±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial 3/		End-of life 3/		Initial 3/		End-of-life 4/		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1-A6	Integrity	II	S	10	∞	10	∞	---	---	---	---	MΩ
B	Continuity	III	S	0	1	0	1	---	---	---	---	Ω
C	Output frequency	IV	X	9595.2	9604.8	9592.8	9607.2	9595.2	9604.2	9590.4	9609.6	Hz
D	Output amplitude (initial adjustment)	V	X	4.97	5.03	---	---	---	---	---	---	Vp-p
E	Output amplitude (after initial adjustment)	VI	X	4.95	5.05	4 925	5.075	4.9	5.1	4.9	5.1	Vp-p
F	Output amplitude regulation	VII	S	4.95	5.05	4.925	5.075	4.9	5.1	4.9	5.1	Vp-p
G	Transition time (t_{TLH} and t_{THL})	VIII	X	0.0	1.0	0.0	1.1	0.0	1.0	0.0	1.2	µs
H	Output pulse, overshoot and undershoot	IX	S	0.00	0.25	0.00	0.25	0.00	0.25	0.00	0.25	V
J	Output pulse, droop	X	S	0.00	0.05	0.00	0.05	0.00	0.05	0.00	0.05	V
K	Power dissipation check	XI	S	---	---	---	---	0	1.25	0	1.25	W

1/ "X" parameters shall be 100 percent tested during module acceptance tests.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ Power supply ±1 percent.

4/ Power supply ±15 percent.

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4.4.2 Order of testing. Test conditions A through K shall be performed in any order with the following exceptions. If test condition A is a part of a sequence of tests, the order of testing shall be as follows: A where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through K in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions A through K shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall not be operated. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column, and test condition shall be performed using the initial limits at 25° ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests A through K shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests A through K shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Cross reference of classification. Military Specification MIL-M-28787/132 supersedes Department of the Navy, Naval Ordance Systems Command Drawing No. 10001 - 2658726C dated 7 November 1968. Part number M28787/132-1 supersedes part numbers 2658726 and M28787/132. These parts are interchangeable.

6.4 Module description. The following electrical parameters reflect the 0° to 60°C end-of-life limits from table I.

Maximum power dissipation at V_{CC} = 28.75 V	1.25 W
Supply voltage	25 Vdc ±20%
Output drive load	10 kΩ ±5%

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6.5 Module function. The module consists of one 9600 Hz square wave oscillator.

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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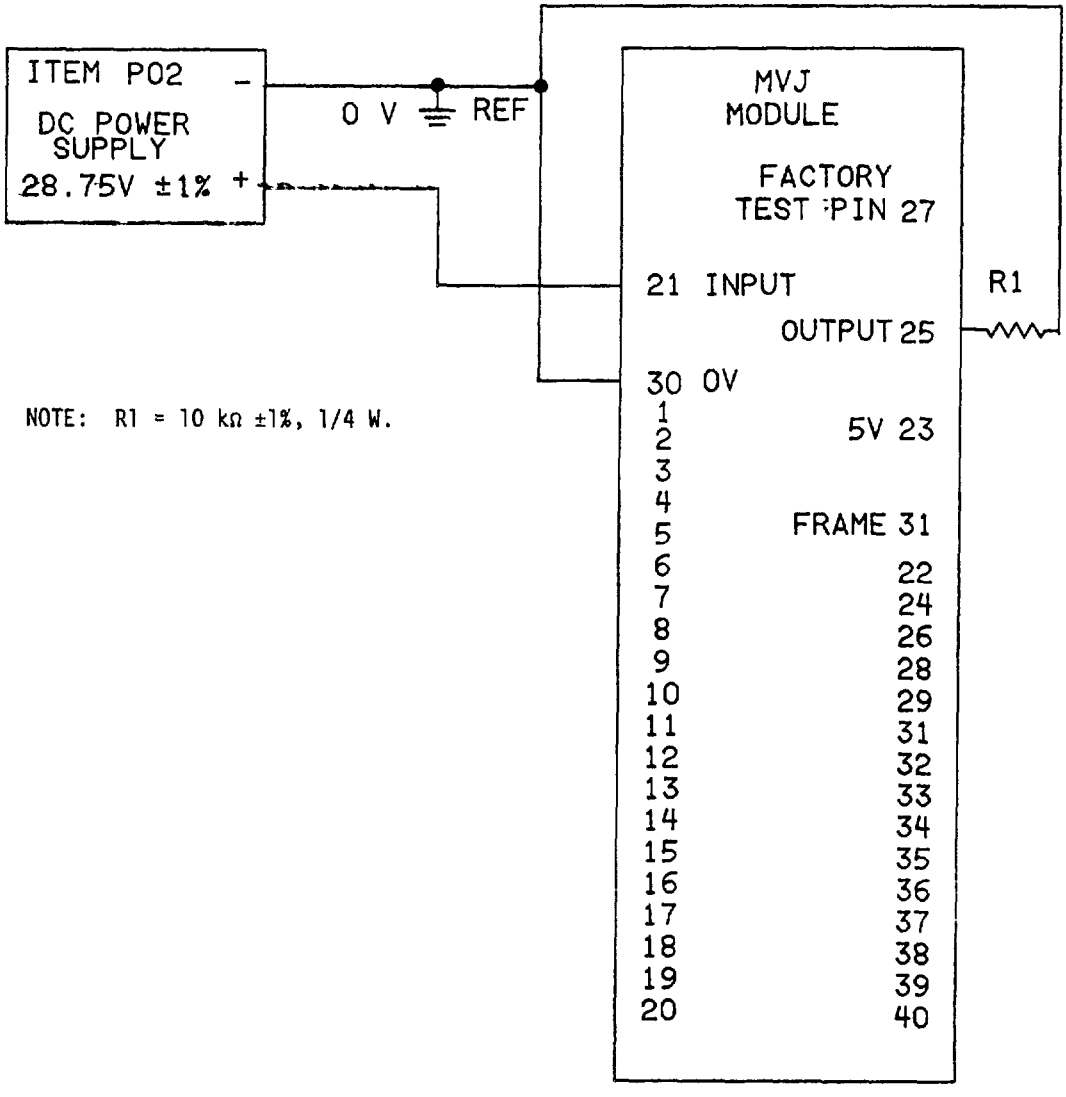


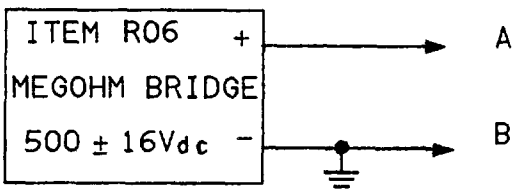
FIGURE 2. Operating and component temperature and life test circuit.

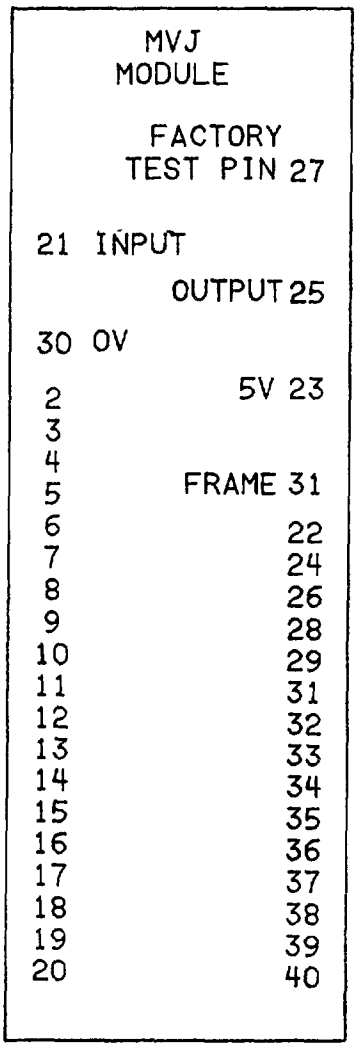
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TABLE II. Integrity test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A1	Integrity	1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20	21,22,23,24,25,26,27,28,29,30,31,32,33,34,35,36,37,38,39,40	10	∞	MΩ	3
A2		1,2,3,4,5,6,7,8,9,10,21,22,23,24,25,26,27,28,29,30	11,12,13,14,15,16,17,18,19,20,31,32,33,34,35,36,37,38,39,40				
A3		6,7,8,9,10,21,23,25,27,30,36,37,38,39,40	1,2,3,4,5,11,12,13,14,15,22,24,26,28,29,31,32,33,34,35				
A4		1,2,8,9,10,13,14,15,16,17,22,24,33,34,35,36,37	3,4,5,6,7,11,12,18,19,20,26,28,29,31,32,38,39,40				
A5		2,5,7,8,9,12,13,14,17,18,24,29,32,35,36,39,40	1,3,4,6,10,11,15,16,19,20,22,26,28,31,33,34,37,38				
A6		3,8,13,15,19,26,34,39	4,9,14,16,20,28,35,40				

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WARNING

When using the megohm bridge, high voltage may be present on the output binding posts Always switch the bridge to discharge before connecting or disconnecting the test leads.

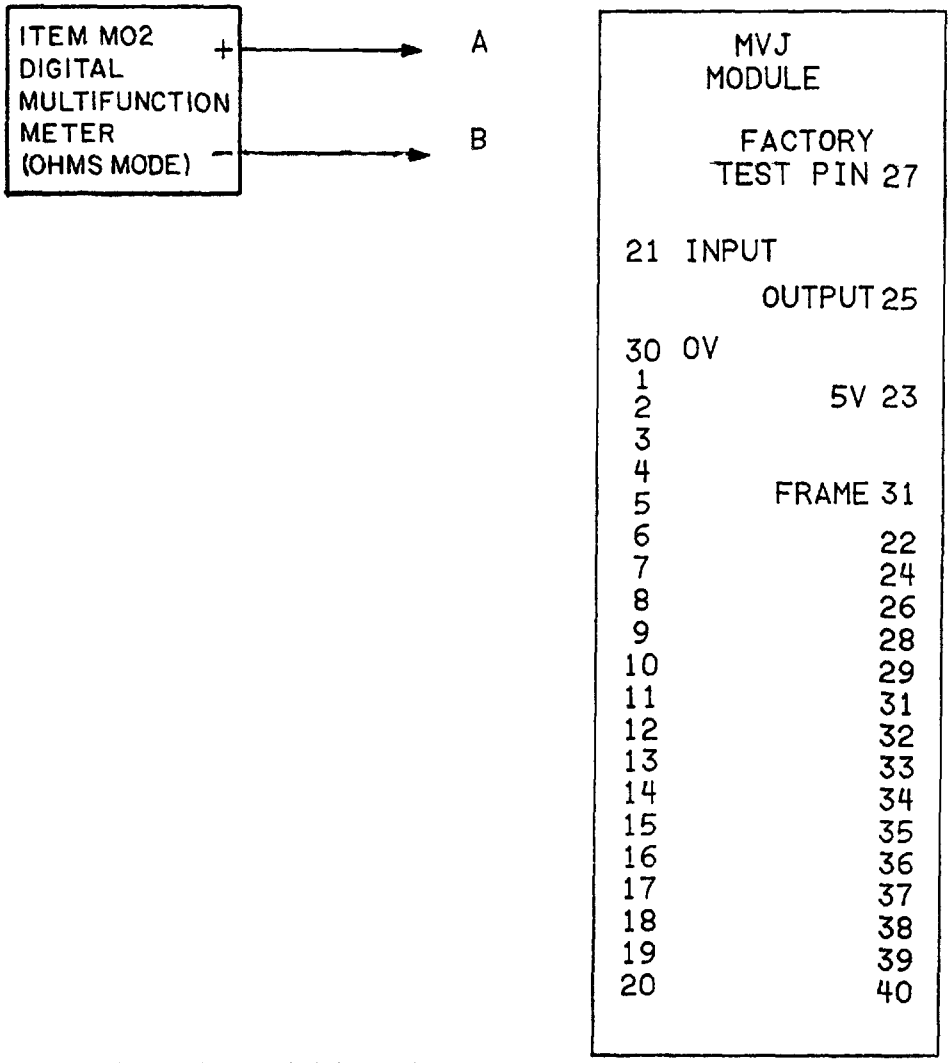
FIGURE 3. Integrity test circuit for table II.

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TABLE III. Continuity test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
B	Continuity	31	Frame	0	1	Ω	4

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NOTE: Penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

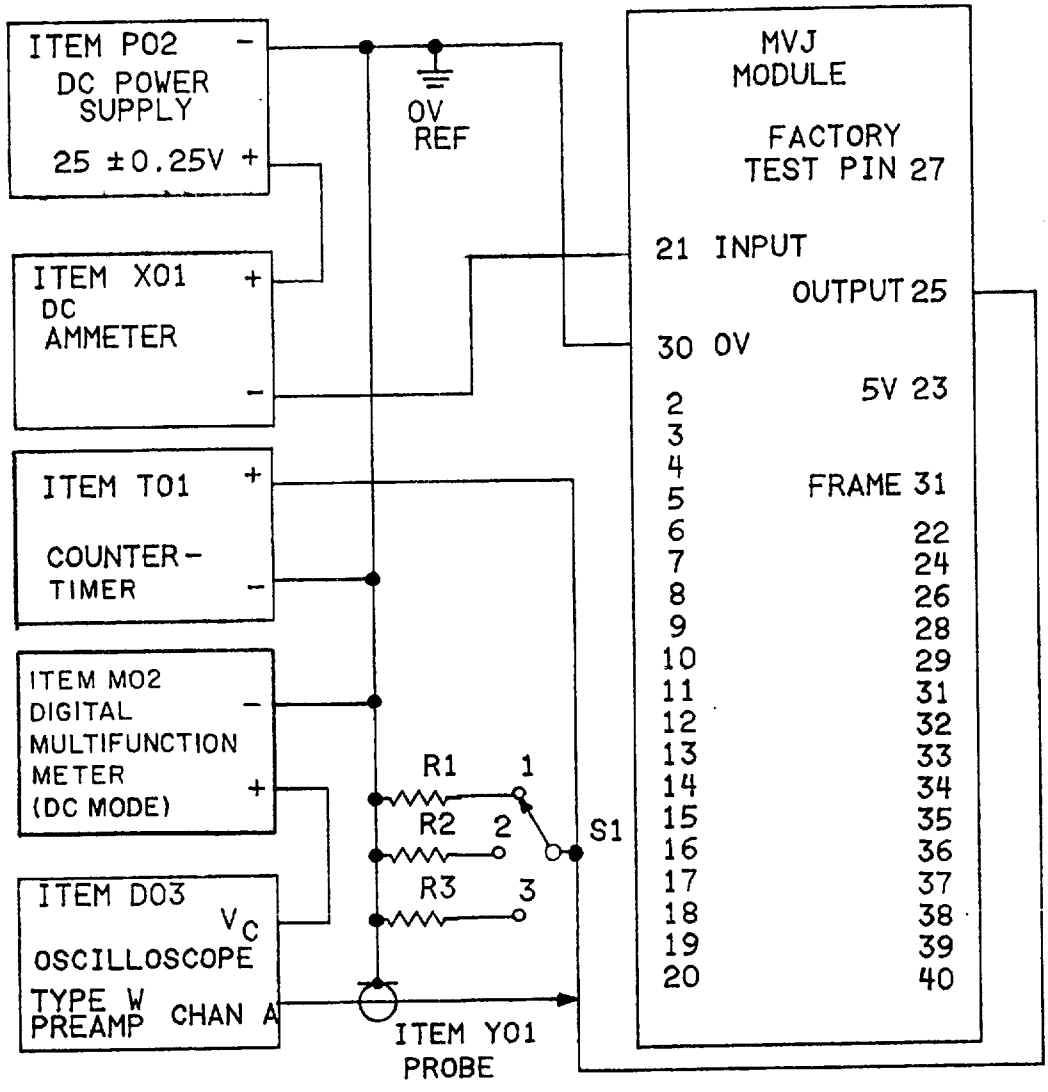
FIGURE 4. Continuity test circuit for table III.

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TABLE IV. Output frequency test.

Test condition	Parameter	Switch S1 to position	Initial limits 25°±5°C ambient		Unit	Test figure
			Min	Max		
C	Output frequency	1	9595.2	9604.8	Hz	5

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CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

NOTES:
1. R1 = 9.53 k Ω ±0.1%, 1/4W.
2. R2 = 10 k Ω ±1%, 1/4W.
3. R3 = 10.5 k Ω ±.1%, 1/4W.
4. S1 = SP 3P switch. Set switch to position 1.

FIGURE 5. Output frequency test circuit for table IV.

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TABLE V. Output amplitude (initial adjustment) test.

Test condition	Parameter	Initial limits 25°±5°C ambient		Unit	Test figure
		Min	Max		
D	Output amplitude (initial adjustment)	4.97	5.03	Vp-p	6

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TABLE VI. Output amplitude (after initial adjustment) test.

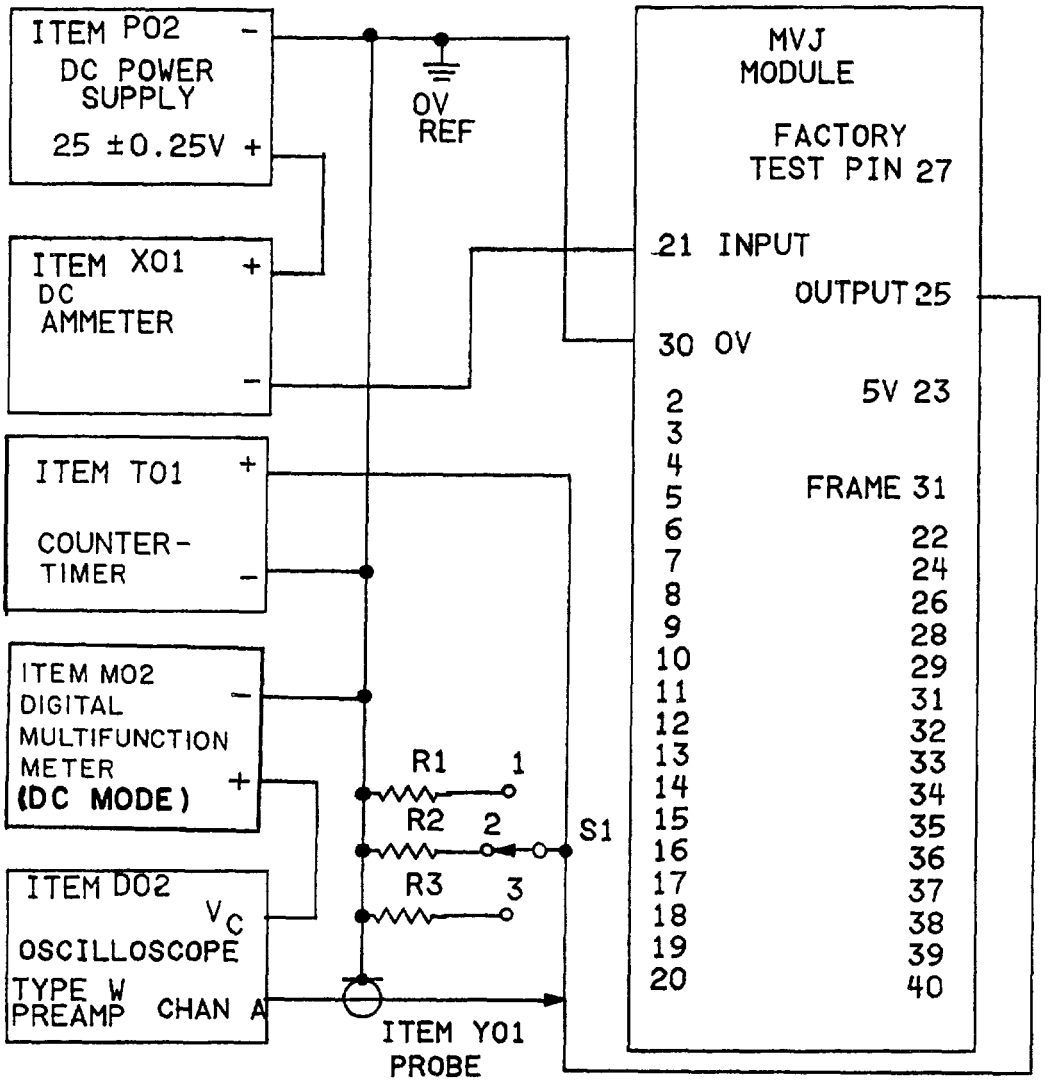
Test condition	Parameter	Initial limits 25°±5°C ambient		Unit	Test figure
		Min	Max		
E	Output amplitude (after initial adjustment)	4.95	5.05	Vp-p	6

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TABLE VII. Output amplitude regulation test.

Test condition	Parameter	Initial limits 25°±5°C ambient		Unit	Test figure
		Min	Max		
F	Output amplitude regulation	4.95	5.05	Vp-p	6

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CAUTION:
Do not isolate the chassis of ammeter X01 from earth ground.

FIGURE 6. Output amplitude and output amplitude regulation test circuit for tables V, VI, and VII.

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NOTES: 1. R1 = 9.53 kΩ ±0.1%, 1/4 W.

2. R2 = 10 kΩ ±1%, 1/4 W.

3. R3 = 10 5 kΩ ±0.1%, 1/4 W.

4. S1 = SP3P switch.

5. Equipment calibration. Before beginning the output voltage test, the test equipment shall be calibrated using the following procedure.

   (a) Set up the module as shown, with switch S1 in position 2. Calibrate item D02 preamp by grounding input chan A and setting the V_C voltage switch to zero. With the mV/cm switch in the 10 mV/cm setting, adjust the vertical position knob for a trace which coincides with the center of the oscilloscope face.

   (b) Set the display control to A-V_C and set the V_C range switch to plus 11 volts. Adjust the V_C comparison dial until item M02 reads 2 500 ± 0.001 V.

6 Test procedure for output amplitude adjustments·

   (a) Apply power to the module. NOTE: For module retest, omit (b) through (d).

   (b) Adjust the module trim pot (figure 7). located on the top of the module, until the center of the positive waveform coincides with the previously calibrated center of the graticule plus or minus 0.25 centimeter on item D02. This corresponds to a positive peak of 2.500 ± 0.0025 V.

   (c) Next, switch the V_C range switch to the minus 11 volt position and readjust the comparison voltage control until the center of the negative position of the waveform coincides with the reference line plus or minus 0.25 centimeter.

   (d) Read item M02. The voltage shall not exceed - 2.500 ± 0.025 V. When conditions as specified in (a) and (d) have been met, the requirements as specified in table V, test condition D will be satisfied. (Once calibration has been completed, do not alter the position of the trimpot for the remainder of the tests). NOTE. For initial acceptances tests, omit (e) through (g).

   (e) Adjust the V_C comparison dial until the center of the positive waveform coincides with the previously calibrated center of the graticule ± 0.25cm on item D02.

FIGURE 6. Output amplitude and output amplitude regulation test circuit for tables V, VI, and VII- Continued.

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(f) Read item M02 and record the reading.

(g) Repeat step (c). Read item M02 and add the reading recorded in step (f). The sum shall comply with table VI.

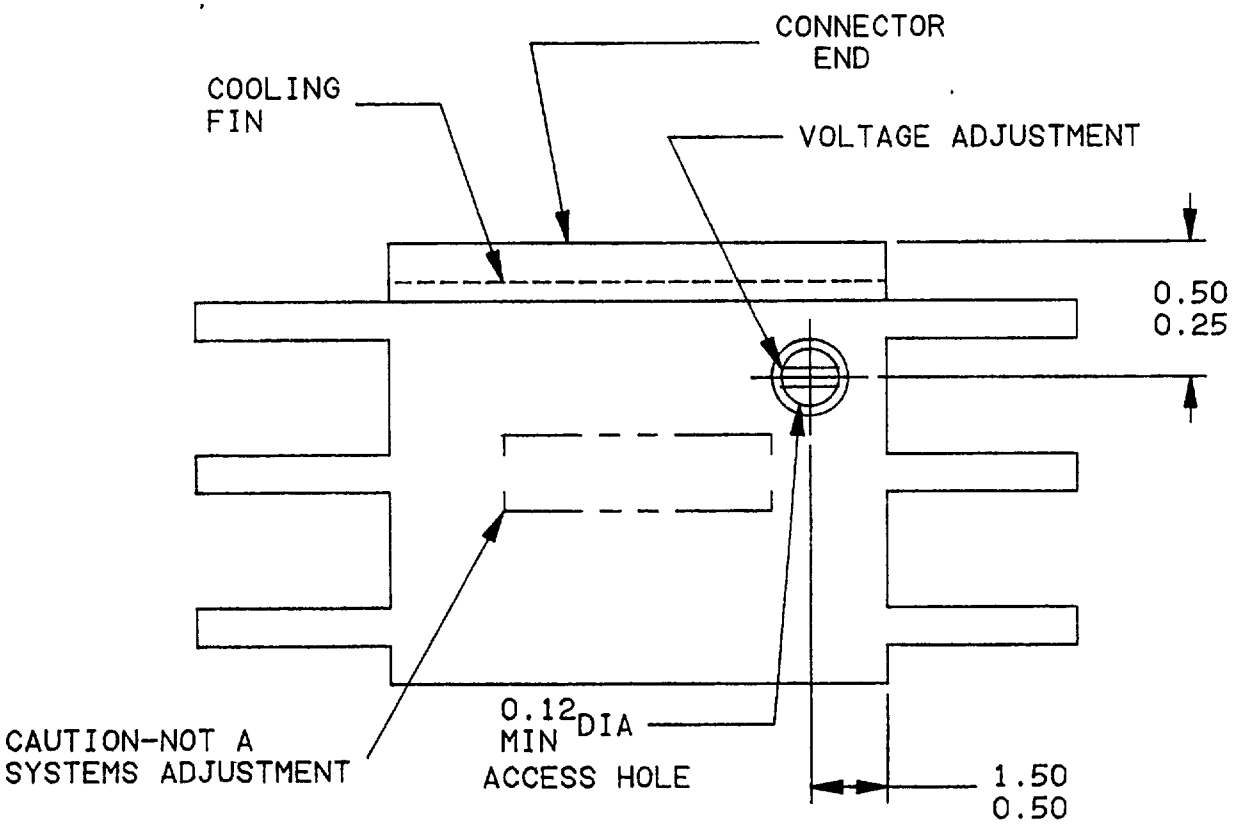
7. Test procedure for output amplitude regulation test:

(a) With the test circuit as shown on figure 6 and with switch S1 in position 1, repeat the steps in Note 6, as applicable. The sum of the item M02 readings obtained in Note 6(g) shall comply with table VII.

(b) Repeat (a) with switch S1 in position 3. The sum of the item M02 readings obtained in note 6(g) shall comply with table VII

FIGURE 6. Output amplitude and output amplitude regulation test circuit for tables V, VI, and VII - Continued.

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INCHES	mm
0.12	3.05
0.25	6.35
0.50	12.7
1.50	38.1

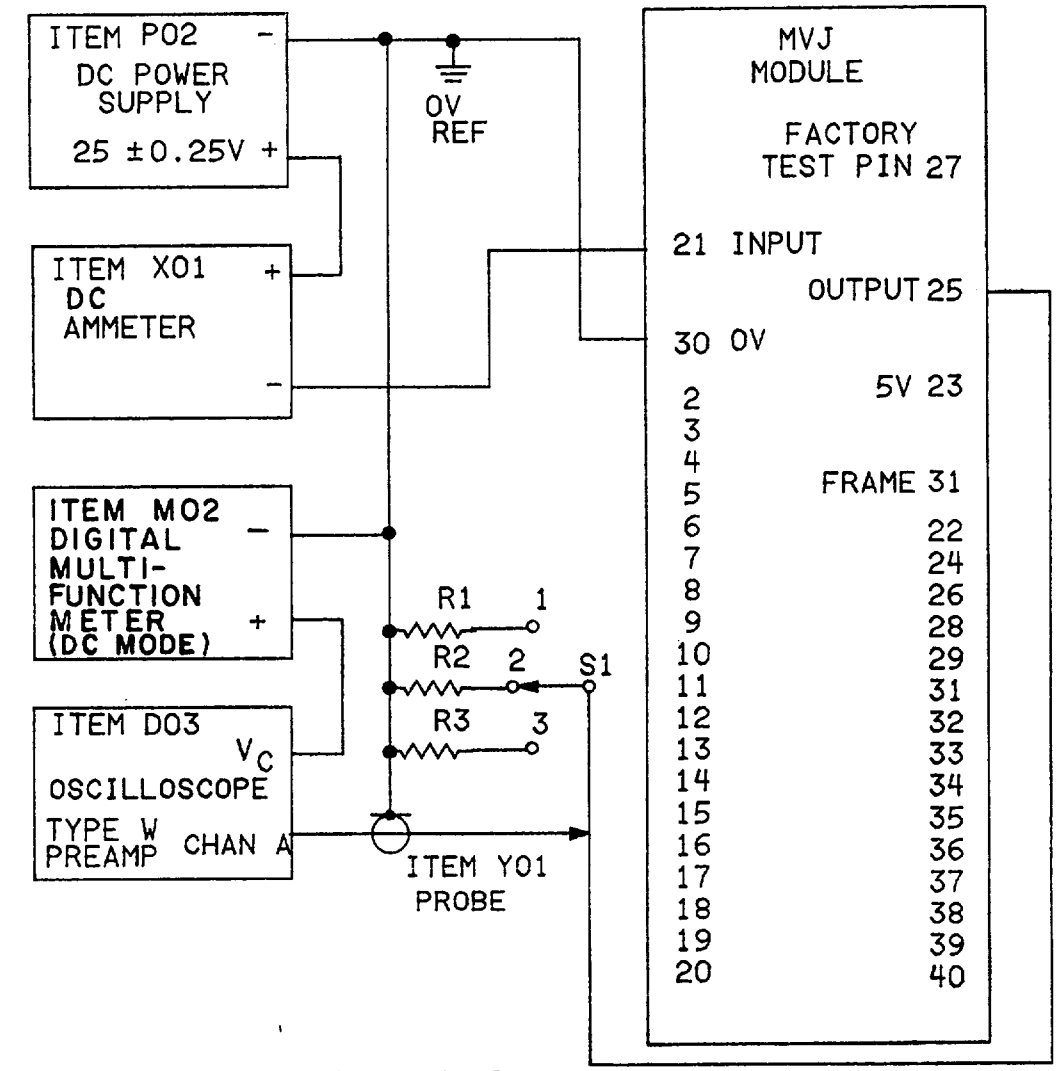
FIGURE 7. Location of screwdriver voltage adjustment.

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TABLE VIII. Transition time (t_{TLH} and t_{THL}) test.

Test condition	Parameter	Initial limits 25°±5°C ambient		Unit	Test figure
		Min	Max		
G	Transition time (t_{TLH} and t_{THL})	0.0	1.0	µs	8,10

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CAUTION: Do not isolate the chassis of ammeter X01 from earth ground.

NOTES:
1. R1 = 9.53 kΩ ±0.1%, 1/4 W.
2. R2 = 10 kΩ ±1%, 1/4 W.
3. R3 = 10.5 kΩ ±0.1%, 1/4 W.
4. S1 = SP3P switch.
5. Set switch S1 to position 2.

FIGURE 8. Transition time (t_{TLH} and t_{THL}) test circuit for table VIII.

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TABLE IX. Output pulse (overshoot and undershoot) test.

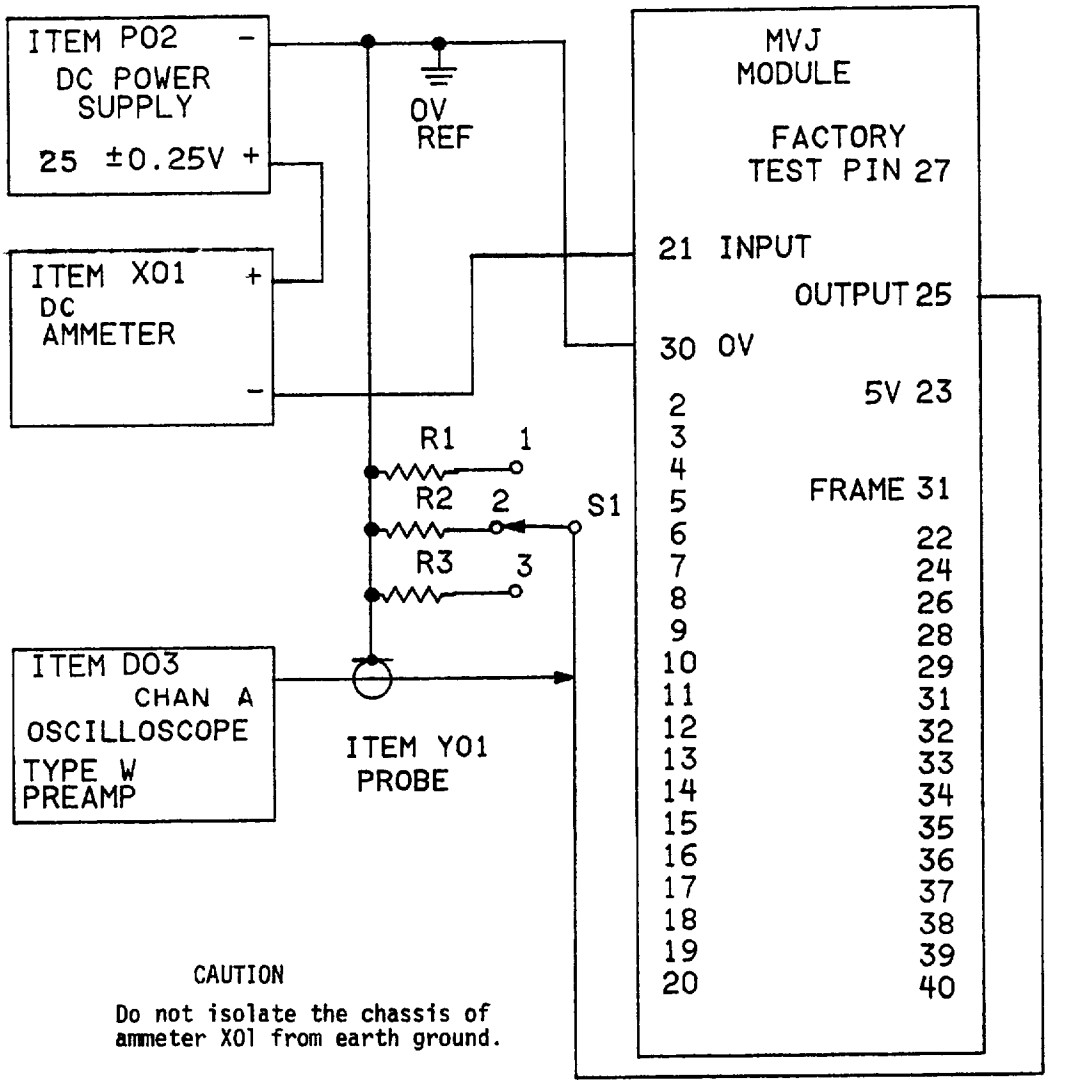
Test condition	Parameter	Initial limits 25°±5°C ambient		Unit	Test figure
		Min	Max		
H	Output pulse, overshoot and undershoot	0.00	0.25	V	9,10

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TABLE X. Output pulse (droop) test.

Test condition	Parameter	Initial limits 25°±5°C ambient		Unit	Test figure
		Min	Max		
J	Output pulse, droop	0.00	0.05	V	9,10

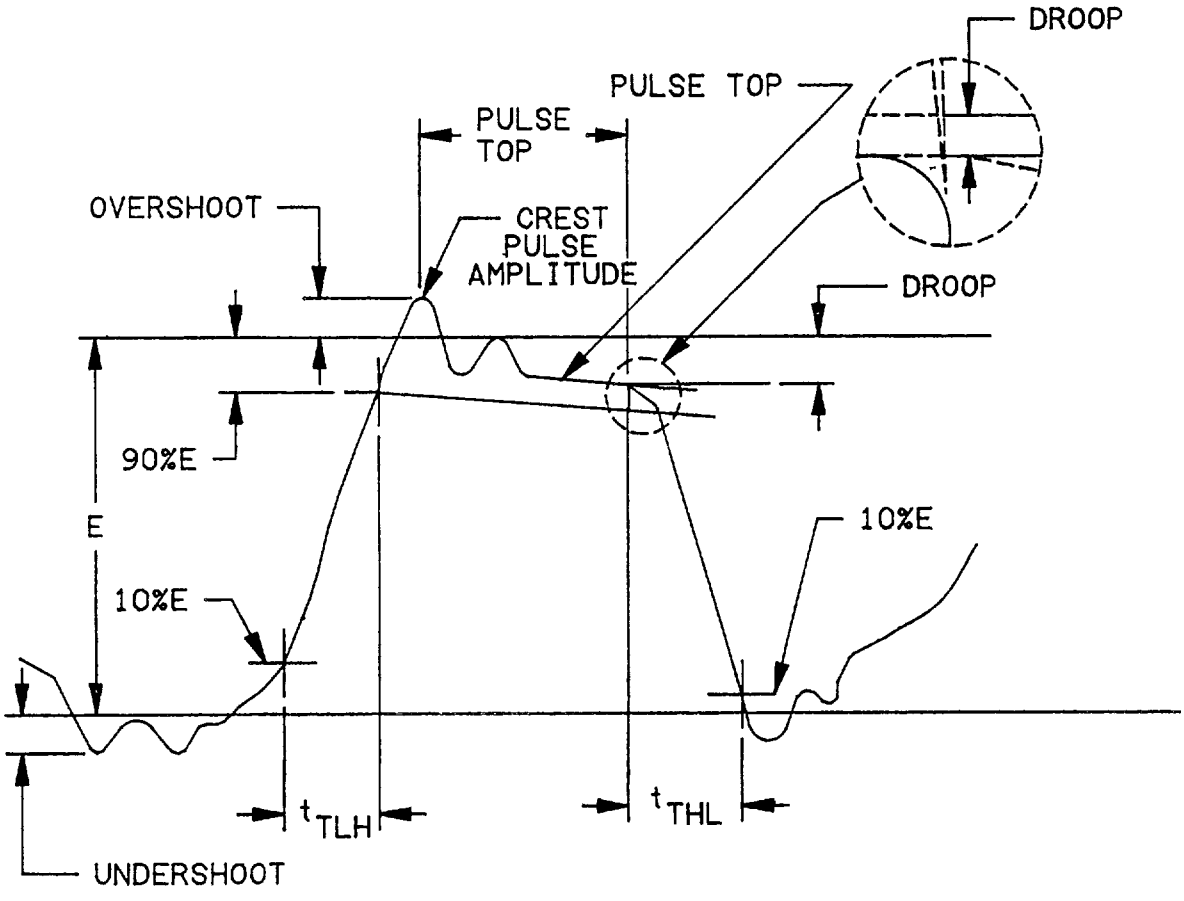
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NOTES:
1. R1 = 9.53 kΩ ±0.1%, 1/4 W.
2. R2 = 10 kΩ ±1%, 1/4 W.
3. R3 = 10.5 kΩ ±0.1%, 1/4 W.
4. S1 = SP3P switch.

FIGURE 9. Output pulse (overshoot, undershoot and droop) test circuit for tables IX and X.

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NOTE: E=PEAK PULSE AMPLITUDE

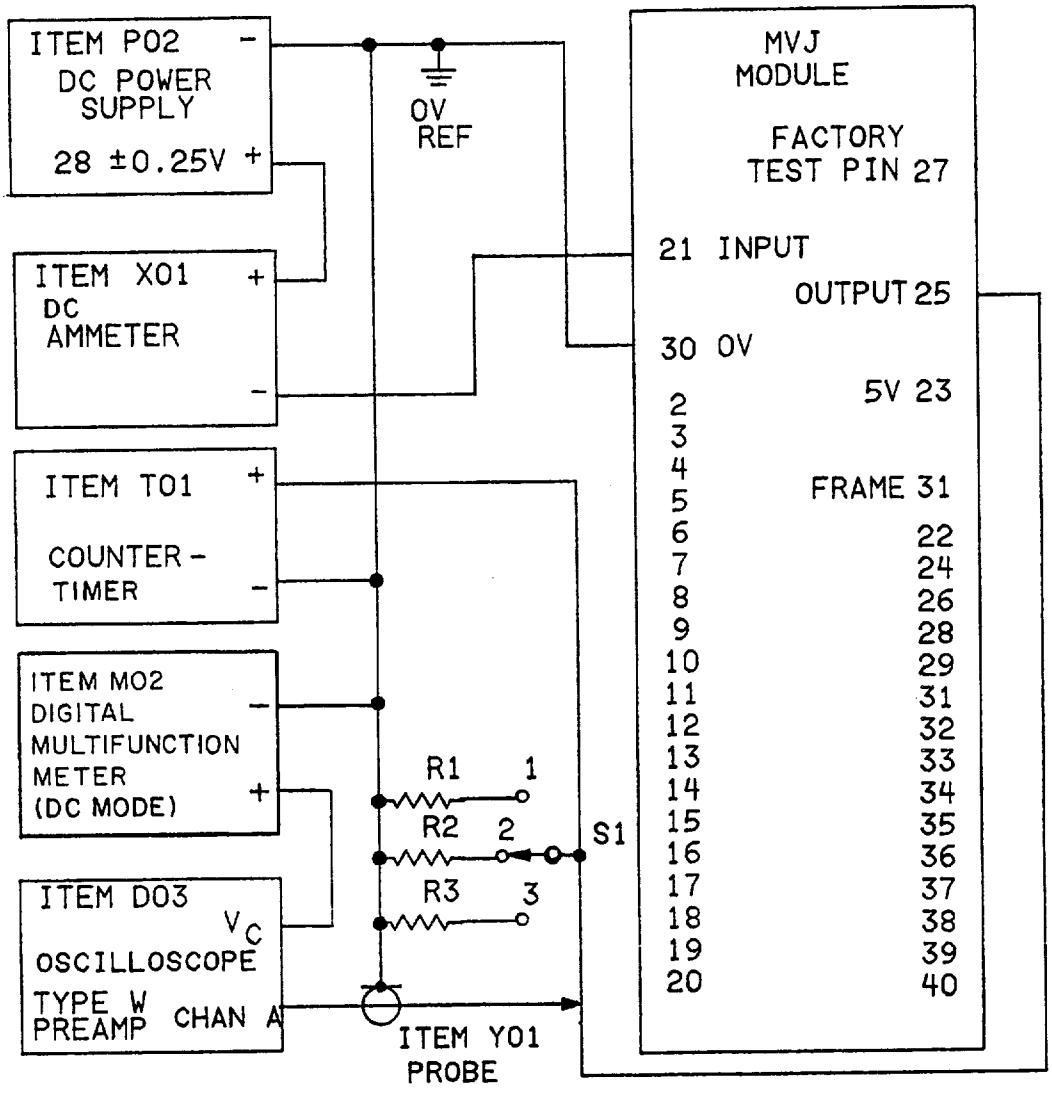
FIGURE 10. Output pulse for tables VIII, IX and X.

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TABLE XI. Power dissipation check test.

Test condition	Parameter	Initial limits 25°±5°C ambient		Unit	Test figure
		Min	Max		
K	Power dissipation check	0	43.6	mA	11

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CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

NOTES:
1. R1 = 9.53 kΩ ±0.1%, 1/4 W.
2. R2 = 10 kΩ ±1%, 1/4 W.
3. R3 = 10.5 kΩ ±0.1%, 1/4 W.
4. S1 = SP3P switch
5. Set switch S1 to position 2.

FIGURE 11. Power dissipation check test circuit for table XI.

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